TECHNICAL BLUEPRINT — V2.1, RED-TEAM HARDENED

Wallace Ad Intelligence Engine

The complete mathematics, gates, state machines, and fail-closed reconciliation logic behind the advertising decision system — including every estimator, every bound, every guard against a model flattering its own recommendations, and the seven-round adversarial review that shaped them.

EMPIRICAL BAYES · WILSON BOUNDS · FRÉCHET OVERLAP BOUNDS · FAIL-CLOSED JOINS · IMMUTABLE BASELINES

Wallace Real Estate • August 2026 • Private — strategy repository: Wallace_Adwerx

1 System architecture

The engine is two scheduled n8n workflows plus a private GitHub repository acting as the durable, auditable store. Everything is dry-run: no code path exists that can create, modify, or fund a live campaign.

COMPONENT	CADENCE	RESPONSIBILITY
Weekly Recommendations `xeRcOu0G5hkaRzmk`	Weekly	Signal scouting (GA4, GSC, SEMrush MCP), QAV computation, gates, opportunity scoring, portfolio allocation, execution kits (OpenAI), brief + dashboard + registry commits
Campaign Tracker `gt7U4Oiy9JXC2B5m`	Friday	Registry left-join, exact-ID spend/delivery reconciliation, day-7/30/60/90 verdicts, scorecard commits
Spend Reader `UoBqxOyBcD2E74AK`	Event	Parses forwarded Adwerx receipts/reports; fail-closed (UNVERIFIED SPEND on any ambiguity)
Failure SMS `lQ46rSaBahKirK69`	Event	Outbound Twilio alerts. Inbound SMS control is permanently disabled (unauthenticated channel).
Repo: Wallace_Adwerx	—	`data/registry.json` (authoritative), `baselines/` (immutable), `briefs/` , `scorecards/` , `docs/` , `tests/`

> Registry writes use SHA compare-and-swap against GitHub: read SHA → validate schema → write with expected SHA → retry on conflict. One writer per run; no lost updates.

2 QAV: Qualified Action Value with Fréchet overlap bounds

GA4 reports per-event session rates, but not the joint distribution of events within sessions. Summing event values overcounts whenever one session fires several events. QAV therefore reports a **bounded interval** instead of a point estimate.

Weights (business value of a session exhibiting the event):

$$w = \{\textbf{Contact Form} : 1.0,\ \textbf{Listing Conversion} : 1.0,\ \textbf{Custom Form} : 0.6,\ \textbf{Registration} : 0.35,\ \textbf{Click-to-Call} : 0.25\}$$

Let e_k be event-sessions of type k (session rate × sessions, capped at total sessions S), indexed in descending weight order. Each session is credited only its *highest-value* event:

$$\mathrm{QAV}_{\mathrm{low}} = \sum_k w_k \cdot \max\Big(0,\ e_k - \max_{j<k} e_j\Big) \qquad \text{(maximum overlap: event sets nested)}$$

$$\mathrm{QAV}_{\mathrm{high}} = \sum_k w_k \cdot \min\Big(e_k,\ S - \textstyle\sum_{j<k} n_j\Big) \qquad \text{(zero overlap, capped at } S\text{)}$$

These are the Fréchet-style sharp bounds on total session value given only the marginals (n_j = sessions already allocated to higher-value events). Scores use $\mathrm{QAV}_{\mathrm{low}}$; a wide $[\mathrm{QAV}_{\mathrm{low}}, \mathrm{QAV}_{\mathrm{high}}]$ interval reduces Evidence Confidence.

> **Unit-tested.** Synthetic page, $S = 100$, $e_k = 10$ for all five events: engine returns $\mathrm{QAV}_{\mathrm{low}} = 10.0$, $\mathrm{QAV}_{\mathrm{high}} = 32.0$ exactly. The red-team predicted the original "value-priority" loop actually computed the upper bound — it did (returned 32). The fix and the regression test live at `tests/qav_overlap_test.js` . This single correction dropped the flagship candidate from 80/A to 67/D.

3 Empirical-Bayes shrinkage — forecasting only, never evidence

Small pages have noisy conversion rates, so per-page rates are shrunk toward the site rate with a pseudo-count prior $\beta = 150$ sessions:

$$\hat{r}_{\text{page}} = \frac{\text{QAV}_{\text{low}} + \beta\, r_{\text{site}}}{S + \beta}$$

> **The zero-observation rule.** V2.0 let this prior mint points: pages with *zero* measured events earned 13/30 propensity points purely from $\beta\, r_{\text{site}}$ — V1's "neutral point" inflation reborn through different math. The rule is now absolute: `if (observed tier QAV == 0) tier points = 0`. Shrinkage informs forecasts; it can never manufacture evidence.

4 Tier scoring: Wilson lower bound × count ceiling

Conversion propensity (30 pts) splits into hard (Contact_Form + Listing_Conversion, 18), mid (Custom_Form + User_Registration, 9), and soft (Click_To_Call, 3) tiers. A rare site baseline made one observed contact form saturate the entire 18-point hard tier. The fix scores the **one-sided 80% Wilson lower confidence bound** of the shrunk rate, expressed as lift over the site tier baseline, then applies an observed-count ceiling:

$$p = \min\left(1, \frac{v + \beta\, r_{\text{tier}}}{n}\right), \qquad n = S + \beta, \qquad z = 1.282$$

$$p_{\text{lower}} = \max\left(0, \frac{p + \frac{z^2}{2n} - z\sqrt{\frac{p(1-p)}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}\right)$$

$$\text{lift} = \max\left(0, \frac{p_{\text{lower}}}{r_{\text{tier}}} - 1\right), \qquad \text{pts} = \min\Big(\max_{\text{tier}} \cdot \min(1, \text{lift}), \;\; \max_{\text{tier}} \cdot \min\big(1, \tfrac{\text{obs}}{3}\big)\Big)$$

OBSERVED EVENT-SESSIONS IN TIER	CEILING (18-PT HARD TIER)
0	0 — zero-observation rule
1	≤ 6 (one third)
2	≤ 12 (two thirds)
3+	Wilson lower bound decides, up to 18

Why both: a minimum-count rule alone creates a cliff at 3; a linear count cap alone ignores exposure (1 event in 20 sessions ≠ 1 in 2,000). The bound handles exposure and uncertainty; the ceiling stops rare baselines from letting a single event saturate. In practice the commercial page's single contact form scored **0/18** — its lower bound never cleared the site baseline.

5 Position-gap curve — paid opportunity, not organic applause

V2.0 gave maximum points to pages ranking #3 organically — exactly where paid media adds the least incremental search opportunity. The curve now peaks where a page is visible enough to prove demand but not yet winning:

AVG GSC POSITION	1–3	>3–5	>5–10	>10–20	>20–40	>40
Gap points (of 15)	0	5	**15**	12	6	2

6 Momentum: absolute scale caps + three-way demand classification

Percentage growth from tiny bases (12→29 clicks: "+142%!") previously earned 20/20. Momentum is now capped by absolute click volume, and — critically — clicks and impressions are classified separately, benchmarked against the sitewide trend:

CURRENT 28D CLICKS	<20	20–49	50–99	100+
Momentum cap (of 20)	5	10	15	20

SIGNAL	LABEL	TREATMENT
Impressions rising absolutely and vs site trend	**TRUE DEMAND GROWTH**	Scale-capped points up to 20
Impressions declining, but less than the site decline	RELATIVE RESILIENCE	Contextual only, standalone max 5; never labeled rising demand
Clicks ↑ while impressions flat/↓	**VISIBILITY-ONLY**	Standalone max 5 (CTR/rank improvement is already scored in position gap — no double credit)

The portfolio-level trend that anchors this comes from summing all GSC pages across matched 28-day windows:

$$\Delta_{\text{site}} = \frac{I_{\text{cur}} - I_{\text{prev}}}{I_{\text{prev}}} = \frac{710{,}279 - 755{,}200}{755{,}200} = -5.9\%, \qquad \frac{351}{433} \text{ pages declining}$$

which currently triggers the brief-level banner **SITEWIDE DEMAND SOFTENING — SEASONALITY UNRESOLVED** (year-over-year and branded/non-branded splits pending).

7 Audience: organic traffic is a proxy, and priced like one

Until Adwerx checkout supplies a real reach forecast, audience evidence is only organic volume — log-scaled and hard-capped at a third of the component:

$$\text{pts} = \min\left(5,\ \left\lfloor 15 \cdot \frac{\log_{10}(\max(V,1))}{3} \right\rceil\right) \quad \text{of } 15, \qquad V = \text{sessions (or } 5 \times \text{GSC clicks)}$$

Full audience points require platform-verified data: Adwerx reach forecast, promised impressions, or audience size recorded at checkout. Uncertainty is never worth more than 5/15.

8 Opportunity score & the funding matrix

$$\text{Score} = \underbrace{P_{\text{hard}} + P_{\text{mid}} + P_{\text{soft}}}_{\text{propensity } \le 30} + \underbrace{M}_{\text{momentum } \le 20} + \underbrace{G}_{\text{position } \le 15} + \underbrace{A}_{\text{audience } \le 15} + \underbrace{R}_{\text{readiness } \le 10} + \underbrace{F}_{\text{fit } \le 10}$$

Two orthogonal grades accompany every score: **Evidence Confidence** (A–D: sample size, QAV interval width, data completeness — with hard ceilings, e.g. <80 sessions can never grade above C) and **Channel Readiness** (has this destination ever actually run on Adwerx?). Funding requires both:

	HIGH CONFIDENCE	LOW CONFIDENCE
High opportunity	**FUND**	CONTROLLED TEST (small budget)
Low opportunity	**REJECT**	WATCH — needs data (never "maybe")

9 Hard gates (pass/fail, reasons kept forever)

GATE	RULE
Landing page loads	HTTP 2xx/3xx on live check; broken pages are FIX FIRST, not fundable
Evergreen destination	Area/community/lifestyle pages only; individual listings excluded upstream — listing traffic rolls up as area *demand* evidence, never as the area page's conversion proof
Destination class (corporate lane)	Brokerage-owned destinations only. Individual agent/team profiles are DENIED for corporate budget (regex on `/agents|team|our-team|staff/`) — prevents brand-search popularity from converting into brokerage-funded personal promotion. Requires an agent-spotlight lane (eligibility, rotation, consent, funding) or agent funding.
Minimum data	≥50 sessions or ≥20 clicks; below that the verdict is "needs data"
Duplication / history	No active campaign on the destination; prior failures surface with reasons
Budget	Fits under lane cap after reservations (§10)

10 Portfolio allocator & proposal lifecycle

Consumer lane: $1,500/mo cap, max 3 concurrent campaigns. Careers lane: $500/mo, one prospecting campaign; retargeting requires ≥500 monthly visits and a real pixel audience. Budget accounting:

$$\text{available} = \text{cap} - \sum_{\text{status}\in\{\text{launched, approved, proposed}\}} b_i, \qquad \text{held} = \text{cap} - \text{committed} - \text{allocated}$$

Held budget stays held — the allocator never force-spends the cap. Lifecycle state machine:

```
proposed --(7-day TTL, revalidated weekly)--> renewed | expired
proposed --human--> approved --(14-day launch deadline)--> launched | APPROVAL STALE
launched --> paused | stopped        any --> rejected | superseded  (permanent history, budget released)
```

Current registry state: WAL-2026-001 (luxury, fallback, $500 reserved), WAL-2026-004 (careers, $200), WAL-2026-007 (commercial pilot, $500 cap). WAL-2026-002 rejected (destination-class), -003/-005/-006 superseded with reasons. Reserved $1,000 of $1,500; $500 held.

11 The operational-validation override (Pilot 1)

The corrected model funds nothing (commercial 35/D, luxury 39/D — both WATCH). Validating the delivery→attribution→verdict chain still requires one live campaign, so the registry schema encodes a **manual QA override that cannot masquerade as a model pick**:

```
{
  "campaignMode": "operational_validation",
  "modelRecommendation": "WATCH",          // the engine's real verdict, preserved forever
  "manualOverride": true,
  "overrideReason": "Validate Adwerx delivery, reporting, attribution and verdict chain",
  "approvedBy": "Howard",
  "maxBudget": 500,
  "excludedFromScoreCalibration": true   // its outcome can never retune weights as if model-selected
}
```

Decision sequence: inspect Adwerx checkout (no purchase) for commercial and the luxury fallback → record real price, promised impressions, frequency, audience, channel coverage → choose commercial unless economics are materially worse → one campaign, $500 hard cap, signed fair-housing preflight. Success is defined operationally: ID reconciliation, impression reconciliation, exact-UTM attribution, non-duplicated snapshots, and a correct day-7 delivery-vs-tracking-vs-engagement diagnosis. ROI is explicitly secondary.

12 Tracker: exact joins, fail-closed spend, immutable baselines

- **Exact joins only**: native Adwerx campaign ID → WAL registry ID (`utm_id` → GA4 `sessionManualCampaignId`) → exact `utm_campaign`. No fuzzy matching; unmatched spend becomes **UNVERIFIED SPEND** and is excluded from every verdict until a human resolves it.
- **Snapshot semantics**: cumulative delivery reports *replace* the prior cumulative snapshot; overlapping periods are never blindly summed. Web-only reports cannot support total-delivery verdicts (Facebook/Instagram impressions would be silently missing).
- **Immutable baselines**: `baselines/<WAL-id>.md` freezes sessions, QAV interval, tier counts, GSC metrics, score, both grades, creative hash, and exact UTM before approval. Day-7/30/60/90 comparisons run against the frozen file — a moving baseline can never launder a bad outcome.
- **Verdict taxonomy**: delivery failure (not serving) ≠ tracking failure (serving, no attributed sessions) ≠ engagement failure (attributed sessions, no qualified actions) ≠ marketing underperformance. "Insufficient data" is a legitimate verdict.

13 Red-team changelog — seven rounds, every score fell

ROUND	ATTACK FOUND	FIX SHIPPED
1–2	V1 neutral points; raw "leads" overcounting; no registry/lifecycle	QAV weights + shrinkage, registry, fund/test/reject matrix, confidence grades
3	Stale proposals reserving budget forever; concurrent registry writes; insecure inbound SMS	7-day TTL + 14-day approval deadline; SHA compare-and-swap; inbound SMS disabled
4	"Lower bound" QAV was actually the upper bound (predicted return: 32; actual: 32)	Fréchet bounds + unit test; luxury 80/A → 67/D; immutable baselines; native Adwerx ID; destination-class gate (agent profile rejected)
5	$150 "budget" presented without platform evidence; identical 69/C twins undecidable	Checkout-derived budgets only; full evidence cards; disclosure that twin propensity was prior-driven
6	Prior minting 13/30 for zero-QAV pages; max points at position 3; +142% momentum from 12 clicks	Zero-observation rule; position curve inverted; absolute momentum caps; demand vs visibility split. Louisville/Oneida 69 → 34/33
7	One contact form saturating 18/18; 10 unverified audience points; compounded visibility caps	Wilson lower bound × count ceiling (commercial 58 → 35); audience proxy cap 5/15; standalone 5-pt visibility ceiling; sitewide demand flag; operational-validation override schema

Score trajectory under honest math: luxury 80/A → 39/D • commercial 58/D → 35/D • Louisville/Oneida 69/C → 33/30/C. The engine's current recommendation is to fund **nothing** — and that is presented to the board as the system working, not failing.

14 Security & compliance invariants

- No auto-spend: the engine has no write path to any ad platform. Launches are manual, in Adwerx, after a signed preflight.
- Fair housing: named human sign-off per campaign; no protected-class targeting or proxies; geo targeting requires a documented business rationale (HUD guidance).
- Inbound SMS command channel disabled (no signature validation = no control surface). Outbound alerts only.
- Strategy stays in the private repo; public reports never expose it. No credentials in code, logs, or commits.

- Migration workflow `9r2idHmp43p77Mtp` is out of scope and untouched.
- Operational-pilot outcomes are excluded from calibration — selection bias from human overrides can never leak into the model's weights.

15 Validation status

CHECK	RESULT
QAV overlap unit tests (`tests/qav_overlap_test.js`)	**4/4 PASS (low=10, high=32; edge low=40, high=92)**
Weekly workflow dry run (exec 223)	**success, zero errors**
Tracker dry run (exec 224)	**success, zero errors**
Registry JSON schema + lifecycle validation	**valid; reservations $1,000/$1,500, $500 held**
Money spent / campaigns launched	**$0 / none — awaiting human checkout inspection + signed preflight**